

# PAPER\_623 — UQFF Nine-Dimensional Wolfram Force-Triad Projection

*Unified Quantum Field Framework (UQFF)*

Daniel T. Murphy • UQFF Research Consortium • 2026

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**Class:** `UQFFNineDimensionalWolframForceTroadProjectionCalculator` **Number:** #210  
**Source:** grok\_share\_6322ac199.txt (Session 161) **Filed:** Session 161 v5.18 **VDS/DVP/BH26:** VDS (9D Gaussian field) + DVP (d4–d6 DPM vortex channels)

### §1 Abstract

The UQFF force triad (Ug, Um, Ub) is embedded in a nine-dimensional Wolfram hypergraph where each force occupies dedicated dimensional channels. This 9D projection resolves the force decomposition problem: gravity acts in d1–d3, DPM vortex flux in d4–d6, and buoyancy displacement in d7–d9. The resulting hypergraph evolution generates void pockets with frequencies matching M87, Centaurus A, and cluster X-ray jet observations.

### §2 Dimensional Channel Assignment

| Dimensions | Force Channel                         | UQFF Term                                       | VDS/DVP/BH26 Link   |
|------------|---------------------------------------|-------------------------------------------------|---------------------|
| d1–d3      | Ug defect (radial, angular, magnetic) | $Ug1+Ug2+Ug3+Ug4$                               | VDS series d=1,2,3  |
| d4–d6      | Um DPM vortex flux (north/south)      | $\kappa \cdot (DPMn - DPMs) / (\nabla UA)^{26}$ | DVP junction points |
| d7–d9      | Ub buoyancy gradient (displacement)   | $g \cdot (1 - 1/\nabla UA)$                     | BH26 outflow bias   |

## §3 Hypergraph Rewriting Rule

**Void seed:** 9-arity hyperedge  $e_{\blacksquare} = \{v_{\blacksquare}, v_{\blacksquare}, \dots, v_{\blacksquare}\}$

**Rewriting rule:**

$$R_{\text{Wolfram}}(e) \rightarrow (e_{\blacksquare} \cup \{v_{\text{new}}\}, \quad e_{\blacksquare} \cup \{v_{\text{new}}\})$$

where:

- $e$  splits at midpoint;  $v_{\text{new}}$  inherits the centroid of  $e$
- d7–d9 coordinates of  $v_{\text{new}}$  receive outflow bias +0.5 (Ub channel enrichment)
- New node spawned at each iteration for arity  $\geq 4$

This yields a **branching tree** where d7–d9 coordinates grow monotonically outward — simulating jet propagation driven by Ub buoyancy.

## §4 Nine-Dimensional $\nabla\text{UA}$ Field

The full 9D Gaussian vacuum density series:

$$\nabla\text{UA} = \sum_{d=1}^9 \exp(-(x_d - \mu_d)^2 / 2\sigma_d^2) \cdot \text{FUB}_i$$

Each Gaussian kernel assigns a phase-space density to its dimensional channel. The total  $\nabla\text{UA}$  is the sum across all 9 channels, weighted by the buoyancy integral  $\text{FUB}_i$ .

**Characteristic values:**

- d1–d3 (Ug):  $\mu \approx 0.5$ ,  $\sigma \approx 0.15$ , contribution  $\approx \text{Ug1} + \text{Ug2} + \text{Ug3}$
- d4–d6 (DVP):  $\mu \approx 0.5$ ,  $\sigma \approx 0.12$ , contribution  $\approx \kappa \cdot (\text{DPMn} - \text{DPMs})$
- d7–d9 (BH26):  $\mu \approx 0.5 + \text{bias}$ ,  $\sigma \approx 0.18$ , contribution  $\approx \text{Ub outflow}$

## §5 Projection to 3D Observable Space

From 9D hypergraph coordinates to observable 3D jet coordinates:

$$x_{\text{proj}} = P \cdot x_v, \quad P \in \mathbb{R}^{3 \times 9} \quad (\text{QR-orthogonal projector})$$

Scale factor for M87:  $\text{jet\_length} = 4.6 \times 10^1 \text{ m} \rightarrow$  multiply projected coordinates. Scale factor for CenA:  $\text{jet\_length} = 7.7 \times 10^1 \text{ m}$ .

## §6 Frequency Events from DVP Junctions

At each node split, d4–d6 asymmetry signals a DVP junction:

$f_{\text{event}} = |\nabla U A|^3 \times 10^1 \text{ Hz} \quad (\text{BH26 cubic rebound law})$

Top-5 frequency predictions from 50-iteration run:

- $f_{\text{min}} \approx 1.0 \times 10^1 \text{ Hz}$  (hard X-ray)
- $f_{\text{min}}\text{--}f_{\text{max}}$  scale as cumulative  $|\nabla U A|^3$

## §7 Observational Agreement

| Observable                 | UQFF 9D Prediction                            | Data                  |
|----------------------------|-----------------------------------------------|-----------------------|
| M87 jet polarization flips | DVP junction events in d4–d6                  | 3 EHT 2017–2021 flips |
| CenA VHE knots             | High-arity branching in d4–d6                 | MNRAS 2025 VHE knots  |
| X-ray frequency floor      | $f \approx 5.71 \times 10^1 \text{ Hz}$ (M87) | Chandra/EHT Dec 2025  |

## §8 References

- grok\_share\_6322ac199.txt — BigBang Hypergraph Theory (Session 161, Topics D4, D13–D14)
- VDS Definition: session\_161\_vds\_dvp\_bh26\_references.md §2
- Preceding: PAPER\_622 (#209)